Cross-Lingual Mechanisms in Multilingual Language Models and Reinforcement Learning

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Introduction

Motivation, Problem Statement, and Research Questions

Motivation

- Large language models use the same parameters across languages, but perform worse on low-resource languages.[a]
- It is difficult to identify which part of a model is responsible for a behaviour, and difficult to change that behaviour by editing a specific part.
- This thesis studies the difference between what a model can do and what its training makes it do, in two settings:
  - Cross-lingual learning in multilingual language models.
  - Optimization with competing objectives in multi-label classification.

[a] Conneau et al., "XNLI: Evaluating Cross-lingual Sentence Representations"; Shi et al., "Language Models are Multilingual Chain-of-Thought Reasoners".

Common Observation

- Editing a localized internal component is often less effective than changing the objective the model is trained on.
- The cases where an intervention does not work are also informative about how the model is organized.

Three Sub-Problems

1. Can localized internal components be used to control cross-lingual behaviour? We test whether editing language-specific neurons improves downstream performance, not only the language of generation. (Part 1)
2. How does reinforcement-learning fine-tuning change cross-lingual behaviour, and why does it help some languages but not others? We track a multilingual model during Group Relative Policy Optimization (GRPO)[a] and identify the factor that controls the per-language outcome. (Part 2)
3. How can a model be improved on a chosen subset of labels without reducing performance on the rest? We design an objective that targets the chosen labels while keeping the others stable. (Part 3)

[a] Shao et al., "DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models".

Introduction

Research Questions

Questions Addressed in This Thesis

- Are language-specific neurons useful for cross-lingual transfer? Earlier work shows they can control the language a model generates in.[a] It is not clear whether editing them improves performance on tasks such as NLI and QA for low-resource languages. (Part 1)
- Does a multilingual model process its reasoning in English at the middle layers, and is that what holds back low-resource languages? One hypothesis is that the model works in an English-leaning space at the middle layers.[b] We ask whether this holds during reinforcement-learning fine-tuning, and whether it or some other factor explains why certain languages do not improve. (Part 2)
- Can preference optimization, which was developed for text generation, manage competing objectives in classification? We ask whether the ideas in DPO,[c] CPO,[d] and SimPO,[e] together with a group-robust balancing method,[f] give targeted improvements in multi-label classification. (Part 3)

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[a] Tang et al., "Language-Specific Neurons: The Key to Multilingual Capabilities in Large Language Models"; Kojima et al., "On the Multilingual Ability of Decoder-based Pre-trained Language Models: Finding and Controlling Language-Specific Neurons".

[b] Wendler et al., "Do Llamas Work in English? On the Latent Language of Multilingual Transformers".

[c] Rafailov et al., *Direct Preference Optimization: Your Language Model is Secretly a Reward Model*.

[d] Xu et al., "Contrastive Preference Optimization: Pushing the Boundaries of LLM Performance in Machine Translation".

[e] Meng, Xia, and Chen, "SimPO: Simple Preference Optimization with a Reference-Free Reward".

[f] Ramesh et al., *Group Robust Preference Optimization in Reward-free RLHF*.

Two Methods with the Same Abbreviation

- Two methods in this thesis share the abbreviation GRPO:
  - Part 2: Group Relative Policy Optimization,[a] a reinforcement-learning algorithm.
  - Part 3: Group Robust Preference Optimization,[b] a method from robust optimization used to balance the two label groups in FairPO.
- The full name is written out in each part, and the abbreviation alone is not used to refer to either.

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[a] Shao et al., "DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models".

[b] Ramesh et al., *Group Robust Preference Optimization in Reward-free RLHF*.

Part 1

Language-Specific Neurons Do Not Facilitate Cross-Lingual Transfer

Part 1: Language-Specific Neurons

Definition of a Neuron

Neuron as an Activation Dimension

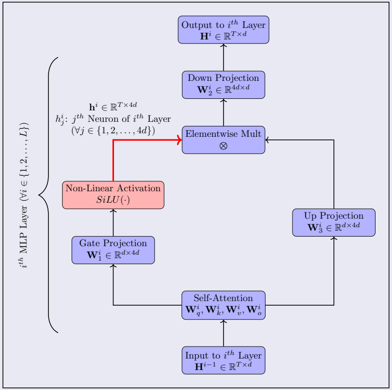
- A neuron is one dimension of the MLP activation vector $\mathbf{h}^i$:

$$\mathbf{h}^i = [\text{act\_fn}(\hat{\mathbf{H}}^i \mathbf{W}_1^i) \otimes (\hat{\mathbf{H}}^i \mathbf{W}_3^i)] \cdot \mathbf{W}_2^i. \quad (1)$$

- Its average activation over token positions, measured on Wikipedia text, is

$$\bar{h}_{i,j} = \frac{1}{T} \sum_{t=1}^{T} h_{i,j}^t. \quad (2)$$

MLP Block in Llama 3



Part 1: Language-Specific Neurons

Methodology: LAPE

Selecting Neurons by Activation Probability Entropy

- A threshold on the Wikipedia activation probabilities is set as

$$\tau = \text{Quantile}_{0.95}\left(p^{l}_{i,j} \mid \forall i, j, l\right). \tag{3}$$

- A neuron (i, j) is a candidate language neuron if

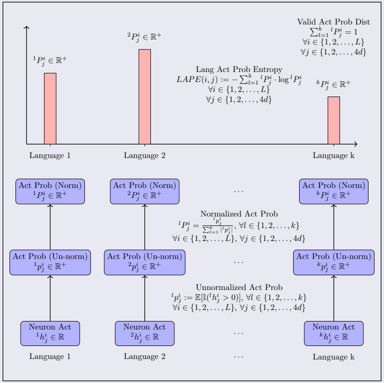
$$\exists\, l \in \{1, \ldots, k\} \text{ such that } p^{l}_{i,j} > \tau. \tag{4}$$

- The 1% of candidates with the lowest LAPE values are kept ($m = 0.01 \times 4d \times L$):

$$LNS = \text{argsort}_m \left\{\text{LAPE}(i, j) \mid \forall i, j,\ \exists l \text{ s.t. } p^{l}_{i,j} > \tau\right\}, \tag{5}$$

$$NTL(i, j) = \left\{l \mid p^{l}_{i,j} > \tau \wedge (i, j) \in LNS\right\}. \tag{6}$$

LAPE Workflow



Methodology: Activation Probability (90p)

Relevance from High-Percentile Activations

- Activation Probability 90p measures how often a neuron's activation exceeds the 90th percentile for its layer, on Wikipedia text.
- For neuron (i, j) and language l:

$$P_q(h_{i,j}^l) = \inf\left\{x \in \mathbb{R} : CDF_{h_{i,j}^l}(x) \geq q\right\}, \quad (7)$$

$$r_{i,j}^l = \mathbb{E}\left[\mathbb{I}\left(h_{i,j}^l > P_{0.9}(h_{i,j}^l)\right)\right]. \quad (8)$$

- Relevance values are collected across all neurons:

$$R^l = \{r_{i,j}^l \mid i \in \{1, \dots, L\},\ j \in \{1, \dots, 4d\}\}. \quad (9)$$

Per-Language Selection

- For each language, keep the top 0.125% neurons by maximum relevance:

$$m = \frac{0.125 \times L \times 4d}{100}, \quad (10)$$

$$LTN(l) = \text{argsort\_desc}_m \left\{r_{i,j}^l \mid \forall i, j\right\}. \quad (11)$$

- This gives the neurons with the highest language-specific relevance for each language.

Part 1: Language-Specific Neurons

Methodology: Neuron-Targeted LoRA

Freezing Selected Neurons during Fine-Tuning

- LoRA fine-tunes a low-rank update to the MLP. The masks are

$$[\text{masked\_A}]_{ij} = 1,\ \forall i \in \{1, \ldots, d\},\ \forall j \in \{1, \ldots, r\}, \quad (12)$$

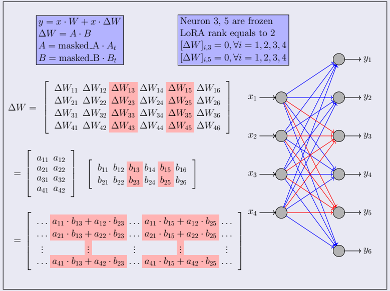
$$[\text{masked\_B}]_{ij} = \begin{cases} 0, & \text{if } j \in \text{Frozen Neurons}, \\ 1, & \text{otherwise.} \end{cases} \quad (13)$$

- Zeroing the columns of B at the frozen neurons stops any update to them:

$$\Delta W_{i,j} = 0, \quad \forall i \in \{1, \ldots, d\},\ \forall j \in \text{Frozen Neurons}. \quad (14)$$

- The update applies to the trainable language neurons in the MLP and to the attention projections (q, k, v, o).

Targeted LoRA Example



Perplexity Change (PPXC)

Measures the effect of an intervention on a target language's perplexity, $PPXC(i,j) = PPX(j \mid \text{set activation to 0 at } i) - PPX(j)$. A larger value means the intervention has a larger effect on the model's handling of language j.

Perplexity Results

Observations

- The aim was to improve cross-lingual performance by intervening on language-specific neurons at test time.
- Editing the activations does not help and often lowers accuracy, which points to neurons with mixed roles.

XNLI Accuracy (%)

Model	Lang	No Int	Int μ	P90	P95	Int 0	P5	P10	P25
Llama 3.1 (LAPE)	vi	80.5	79.5	79.0	78.9	79.8	73.6	77.7	80.5
	hi	75.0	75.2	74.9	75.0	74.4	74.8	75.1	74.9
	ur	70.0	70.4	69.3	69.0	68.5	68.6	68.7	69.7
Llama 3.1 (Act Prob 90p)	vi	80.5	78.2	79.3	79.5	79.0	76.5	77.4	78.0
	hi	75.0	74.1	71.8	71.2	73.7	75.0	74.6	74.2
	ur	70.0	69.7	69.3	68.7	69.6	69.4	69.5	69.5
Mistral Nemo (LAPE)	vi	80.5	80.4	80.6	80.5	79.2	80.6	80.5	80.3
	hi	76.1	69.8	66.9	66.6	74.9	73.0	72.4	71.0
	ur	66.8	66.5	67.0	66.8	66.9	65.8	65.4	65.7
Mistral Nemo (Act Prob 90p)	vi	80.5	67.4	81.1	81.1	79.8	37.4	40.7	47.8
	hi	76.1	72.2	74.5	74.4	74.5	62.4	66.3	69.3
	ur	66.8	65.9	61.3	59.7	66.4	54.6	61.6	65.0

Part 1: Language-Specific Neurons

Results: Test-Time Interventions on XQuAD

Observations

The same edits disrupt task performance on XQuAD, so these neurons take part in task-specific processing and not only language identity.

XQuAD Results (EM, F1 in parentheses)

Model	Lang	No Int	Int μ	P90	P95	Int 0	P5	P10	P25
Llama 3.1 (LAPE)	vi	41(73.5)	40(72.9)	31(69.5)	28(66.8)	32(69.2)	4(35.8)	10(43.2)	39(71.5)
	hi	38(64.1)	40(65.5)	36(65.4)	36(66.3)	23(49.9)	38(62.8)	37(62.8)	39(65.3)
	zh	56(77.5)	10(62.8)	3(56.1)	4(52.9)	33(63.2)	20(49.1)	33(63.2)	22(67.8)
Llama 3.1 (Act Prob 90p)	vi	41(73.6)	39(73.0)	23(64.9)	24(64.8)	42(73.8)	33(67.8)	36(70.3)	39(72.0)
	hi	38(64.1)	34(60.7)	36(62.8)	32(61.9)	38(62.9)	23(54.9)	31(58.6)	35(60.7)
	zh	56(77.5)	61(80.7)	56(78.8)	56(78.4)	55(78.5)	40(61.7)	50(73.6)	56(79.3)
Mistral Nemo (LAPE)	vi	39(74.6)	42(76.8)	40(75.0)	39(73.6)	13(45.0)	10(40.4)	11(41.2)	37(73.3)
	hi	38(66.9)	35(65.9)	37(66.6)	34(66.5)	22(51.7)	36(65.8)	36(66.1)	36(67.1)
	zh	47(74.9)	24(74.0)	0(61.6)	0(59.3)	14(53.3)	1(33.6)	24(68.9)	35(77.0)
Mistral Nemo (Act Prob 90p)	vi	39(74.6)	11(43.3)	29(63.9)	15(48.5)	39(74.5)	0(3.8)	0(6.5)	2(12.9)
	hi	38(66.9)	26(54.4)	37(68.9)	37(68.7)	33(63.8)	0(6.0)	0(11.9)	12(35.2)
	zh	47(74.9)	46(77.4)	20(59.8)	15(54.9)	48(76.2)	0(8.5)	0(17.0)	9(45.6)

Observations

- This tests a longer-lasting intervention: fine-tuning only the language-specific neurons with LoRA.
- The result is not distinguishable from standard LoRA fine-tuning.
- Both test-time edits and targeted fine-tuning fail to help, which supports moving from internal editing to output-level control.

Fine-Tuning Results (Llama 3.1)

FTL	EL	No Int	Int μ	P90	Int 0	P10
en	vi	80.2	79.6	78.5	79.2	78.0
vi	vi	80.1	79.5	78.6	79.2	78.0
en+vi	vi	80.1	79.4	78.5	79.1	78.0
en	hi	74.9	74.6	74.6	74.1	74.6
hi	hi	74.9	74.6	74.5	74.3	74.6
en+hi	hi	74.9	74.5	74.5	74.3	74.7
en	ur	69.8	70.4	69.6	70.2	69.0
ur	ur	69.8	70.5	69.5	70.4	69.1
en+ur	ur	70.0	70.6	69.5	70.3	68.9

Part 1: Language-Specific Neurons

Findings and Takeaway

Summary of Findings

- Direct interventions on language-specific neurons, both at test time and through targeted fine-tuning, do not improve cross-lingual task performance. Where there is a gain it is small and not in a consistent direction, and several interventions lower performance.
- Setting a neuron's activation to zero does not remove its effect, so zero is not a reliable way to turn a neuron off.
- The neurons identified for different languages are not independent, so editing the neurons for one language affects the others.
- These two observations are consistent with neurons that take part in more than one function,[a] which is why surgical edits do not give a clean effect.

[a] Elhage et al., "Toy Models of Superposition".

Publication

Language-Specific Neurons Do Not Facilitate Cross-Lingual Transfer.[a] Accepted at the Sixth Workshop on Insights from Negative Results in NLP, NAACL 2025.

[a] Mondal, Sen, et al., *Language-Specific Neurons Do Not Facilitate Cross-Lingual Transfer*.

Part 2

Cross-Lingual Mechanisms during GRPO Training

Part 2: Cross-Lingual Mechanisms during GRPO Training

Motivation

From a Fixed Model to a Training Procedure

- Part 1 probed a fixed model and found that editing a localized set of neurons does not change task behaviour.
- Here we instead track how a training procedure changes a model's cross-lingual behaviour over time.
- The procedure is Group Relative Policy Optimization (GRPO),[a] used to fine-tune models for reasoning; the behaviour is how the model handles the same math problem in different languages.

[a] Shao et al., "DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models".

Why Reinforcement Learning Raises a Concern

- GRPO rewards the model only for a correct final answer; the verifier does not read the language of the reasoning.
- If the model could raise its reward by leaning more on an English-aligned middle of the stack, training might push it there and widen the gap for low-resource languages.
- Whether GRPO actually changes this internal behaviour, and how the effect varies across languages, is the question.

Part 2: Cross-Lingual Mechanisms during GRPO Training

Background: The Three-Phase Hypothesis

Early-Target, Mid-English, Late-Target

- One account of multilingual transformers:[a] early layers map the input into the model's representation; middle layers work in a space closer to English than to the input language; the last layers map back to the input language.
- A logit-lens readout of intermediate predictions should then show English through the body of the stack and the target language only near the end.

[a] Wendler et al., "Do Llamas Work in English? On the Latent Language of Multilingual Transformers".

What This Means under GRPO

- The hypothesis makes a static prediction we can test directly.
- Tracking it across checkpoints tests whether GRPO changes it: training could lean more on the English-aligned middle, since the reward ignores the language of the reasoning.

Part 2: Cross-Lingual Mechanisms during GRPO Training

Background: GRPO Objective and Analysis Tools

Group Relative Policy Optimization

- An online policy-gradient method:[a] for a prompt, the policy generates a group of G responses, each given a scalar reward R_i^j.
- The advantage standardizes rewards within the group, with no separate value model:

$$A_i^j = \frac{R_i^j - \mu^i}{\sigma^i}, \quad \mu^i = \frac{1}{G}\sum_j R_i^j.$$

- Loss uses a clipped policy ratio with a KL penalty against a frozen reference ($\epsilon = 0.2$, $\beta = 0.04$).

[a] Shao et al., "DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models".

Two Tools for Reading the Model

- Logit lens:[a] apply the final normalization and output head to an intermediate layer, giving a per-layer distribution over the vocabulary. We use it to ask which language the predicted token belongs to at each layer.
- Centred Kernel Alignment:[b] measures similarity between two sets of representations, invariant to rotation and uniform scaling. We use it to compare a language with English, and a checkpoint with the base model.

[a] nostalgebraist, *Interpreting GPT: The Logit Lens*.
[b] Kornblith et al., "Similarity of Neural Network Representations Revisited".

Part 2: Cross-Lingual Mechanisms during GRPO Training

Experimental Setup

Data, Model, and Reward

- MGSM,[a] seven languages: en, bn, te, th, ru, ja, zh. 200 train and 50 eval per language (1,400 / 350 total).
- One-shot prompts; instruction and field headers in English, question and worked example in the target language; final answer as an English integer.
- Policy: Qwen2.5-7B-Instruct ($L = 28$, $d = 3584$); frozen copy as reference.
- Reward is binary, from a verifier combining a regex match with a GPT-4o-mini judge.

[a] Shi et al., "Language Models are Multilingual Chain-of-Thought Reasoners".

Training Settings

Setting	Value
LoRA rank r / α	64 / 16
LoRA targets	q,k,v,o_proj
Group size G	4
Clip ϵ / KL β	0.2 / 0.04
Optimizer	AdamW
Peak LR	1×10^{-5}
Batch size B	8 prompts
Total steps	620
Checkpoints	every 20 (30 total)
Temperature / p	0.7 / 0.95
Max new tokens	512

Part 2: Cross-Lingual Mechanisms during GRPO Training

Tokeniser Coverage

Script Tokens in the Qwen2.5 Vocabulary

Language	Coverage class	Tokens (approx.)
Chinese	High (CJK base)	$\sim 25{,}000$
Japanese	High (kanji, kana)	$\sim 6{,}000$
Russian	High (Cyrillic)	$\sim 4{,}000$
Thai	Mid	$\sim 2{,}500$
Bengali	Low	~ 46
Telugu	Very low	~ 14
English	Bulk of vocab	—

What to Note

- The number of dedicated tokens differs by orders of magnitude across scripts.
- Thai has about 2,500 tokens, Bengali about 46, Telugu about 14.
- This gap lines up with the one language GRPO does not lift, as the results show.

Part 2: Cross-Lingual Mechanisms during GRPO Training

The Five Measurements

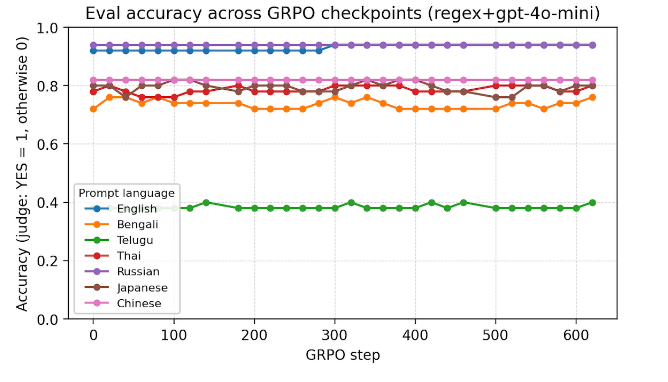
What We Compute at Each Checkpoint

1. End-task accuracy: fraction of evaluation problems solved per language, judged by the verifier.
2. Per-layer language distribution: from the logit lens, how much prediction mass falls on each language at each layer, renormalized over the seven languages.
3. Three-phase analysis: how often the top-1 logit-lens prediction is a given language in the early, middle, and late phases.
4. English-minus-target gap: per layer, the English fraction minus the target fraction of top-1 predictions.
5. Representation geometry: CKA cross-lingual similarity to English, and within-language drift from step 0.

Part 2: Cross-Lingual Mechanisms during GRPO Training

Result: Accuracy over Training

Accuracy across 620 Steps



First and Last Checkpoint

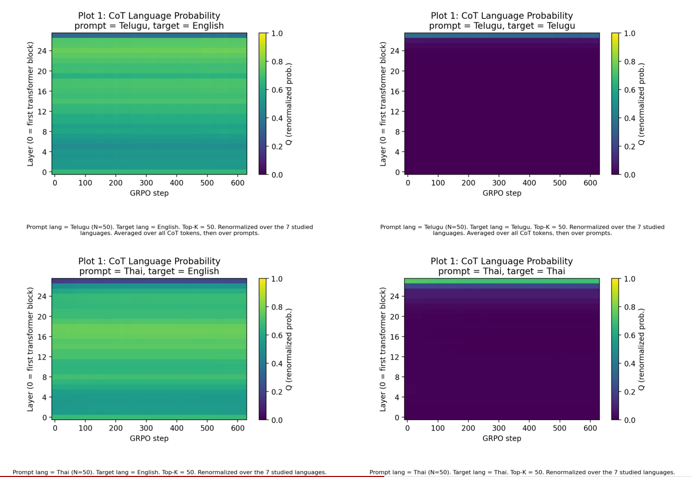
Lang	Step 0	Step 620	Δ
en	0.92	0.94	$+0.02$
ru	0.94	0.94	$+0.00$
zh	0.82	0.82	$+0.00$
ja	0.80	0.80	$+0.00$
th	0.78	0.80	$+0.02$
bn	0.72	0.76	$+0.04$
te	0.38	0.40	$+0.02$

- Six languages stay between about 0.72 and 0.94; Telugu stays at about 0.38 to 0.40.
- GRPO moves accuracy by at most a few points.

Part 2: Cross-Lingual Mechanisms during GRPO Training

Result: Predicted Language at Each Layer

Logit-Lens Language by Layer and Step



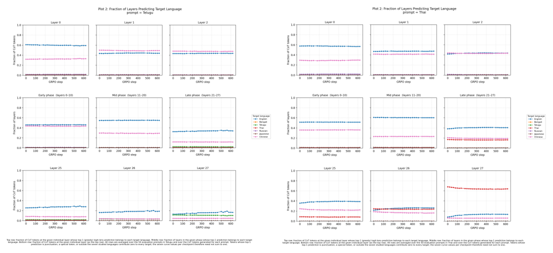
Reading

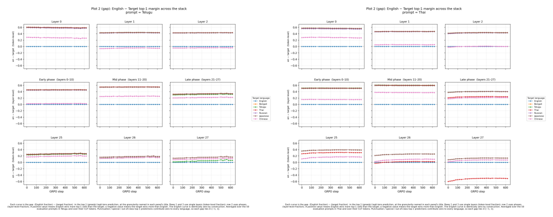
- For a non-English prompt, early and middle layers put most mass on English; the target language appears only in the last two or three layers.
- The pattern is the same across prompt languages.
- It is flat across all 620 steps: GRPO neither builds up nor removes the English-leaning middle.

Part 2: Cross-Lingual Mechanisms during GRPO Training

Result 3: Three-Phase Analysis and the English Lead

Top-1 Language per Phase and the English-Minus-Target Gap



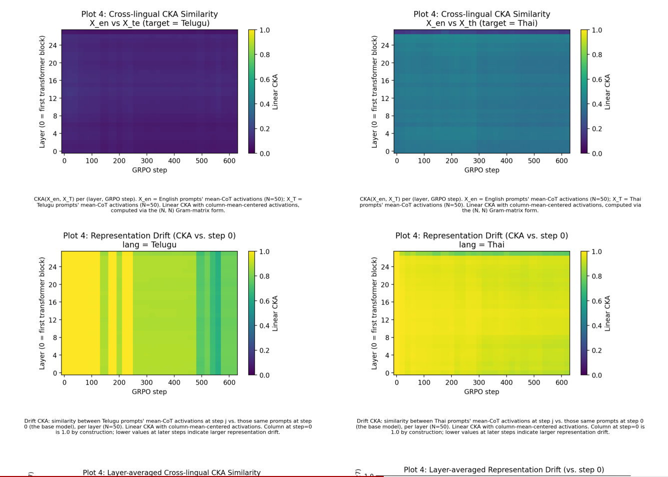


Reading

- English leads the early and middle phases, highest in the middle; the target language appears mainly in the late phase.
- At the last layer, Thai overtakes English while Telugu stays low. The English-minus-target gap is positive through the body of the stack and flat across the 620 steps.

Result: Representation Geometry (CKA)

Cross-Lingual Similarity and Within-Language Drift



Reading

- Similarity to English is flat over training and tracks coverage: Telugu and Bengali lowest (about 0.10 and 0.15), Russian about 0.32, Thai/Japanese/Chinese about 0.40.
- Within-language drift: every language stays above about 0.90, except Telugu, which drops to about 0.63 before settling near 0.78.
- The one language that does not improve is the one whose representations move the most.

GRPO Does Not Cause Cross-Lingual Collapse

- The model reads out as English through the early and middle layers and surfaces the target language only at the end, matching the static three-phase picture.
- This is already present at step 0 and does not change over 620 steps. The per-layer distribution, the three-phase commitment, and the English-minus-target gap are all flat in training.

Tokeniser Coverage and the Telugu Case

- Telugu sits far below the rest, does not improve, and is also the language whose representations move the most.
- With very few dedicated tokens, its text is split into long, coarse byte sequences, so the answer-level reward does not turn into a useful per-token gradient.
- The limiting factor is the granularity of the reward signal, set by coverage, not language identity or a change in internal routing.

Part 3

FairPO: Robust Preference Optimization for Fair Multi-Label Learning

Part 3: FairPO

Motivation: Fair Multi-Label Learning

Where Standard Training Falls Short

Multi-label classification (MLC) is used in content tagging, medical diagnosis support, and skill identification from resumes. Models trained to optimize overall metrics can:

- Do well on frequent labels (non-privileged, $\bar{\mathcal{P}}$) but poorly on rare but important labels (privileged, $\mathcal{P}$).
- Leave a large gap between the two groups: $perf(\mathcal{P}) \ll perf(\bar{\mathcal{P}})$.
- Example: a medical system that keeps missing rare disease labels in $\mathcal{P}$.

Standard losses such as BCE do not account for this, and simple re-weighting does not reliably balance the two groups or handle confusing labels.

What FairPO Aims to Do

- Target the performance gap between the two label groups $\mathcal{P}$ and $\bar{\mathcal{P}}$ directly.
- Use robust optimization to balance the groups, so improving one does not come at the cost of the other.

Part 3: FairPO

Problem Setup and Fairness Goals

Setup, Confusing Sets, and the Two Goals

- MLC with T labels, $\mathcal{T} = \{1, \ldots, T\}$, dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, with $y_{il} = +1$ if label l applies to x_i and 0 otherwise.
- Partition $\mathcal{T}$ into privileged $\mathcal{P}$ and non-privileged $\bar{\mathcal{P}}$. Model score for label t is $m(x_i; \mathbf{w}_t)$; reference score is $m(x_i; \hat{\mathbf{w}}_t)$.
- Confusing sets for a privileged label $l \in \mathcal{P}$:
  - If $y_{il} = +1$: confusing negatives $S_{il}^{\text{neg}} = \{k \mid y_{ik} = 0,\ m(x_i; \mathbf{w}_k) \geq m(x_i; \mathbf{w}_l)\}$.
  - If $y_{il} = 0$: confusing positives $S_{il}^{\text{pos}} = \{k' \mid y_{ik'} = +1,\ m(x_i; \mathbf{w}_{k'}) \leq m(x_i; \mathbf{w}_l)\}$.

  Overall confusing set $S_{il} = S_{il}^{\text{neg}} \cup S_{il}^{\text{pos}}$.
- Fairness goals:
  - Privileged: if $S_{il} \neq \emptyset$, push $m(x_i; \mathbf{w}_l) \gg m(x_i; \mathbf{w}_k)$ for $k \in S_{il}^{\text{neg}}$ and $m(x_i; \mathbf{w}_l) \ll m(x_i; \mathbf{w}_{k'})$ for $k' \in S_{il}^{\text{pos}}$; otherwise use BCE.
  - Non-privileged: keep performance close to the reference, $\ell(\mathbf{w}_j) \leq \ell(\hat{\mathbf{w}}_j) + \epsilon$.

Part 3: FairPO

Preference-Based Privileged Loss

Framing Hard Privileged Cases as Preferences

- For a privileged label l that is truly positive, the model may score a confusing negative $k \in S_{il}^{\text{neg}}$ higher; we want $m(x_i; \mathbf{w}_l) \gg m(x_i; \mathbf{w}_k)$.
- For a privileged label l that is truly negative, the model may score it above an actual positive $k' \in S_{il}^{\text{pos}}$; we want $m(x_i; \mathbf{w}_l) \ll m(x_i; \mathbf{w}_{k'})$.
- Following DPO,[a] we treat these as preferences ($l \succ k$, or $l \prec k'$ in score terms) and use a comparison term measured against a reference. If $S_{il} = \emptyset$, BCE is used for label l.

[a]Rafailov et al., *Direct Preference Optimization: Your Language Model is Secretly a Reward Model*.

Privileged Loss

$$h_{\mathbf{w}}(x_i, l, k) = \left(\log \frac{m(x_i; \mathbf{w}_l)}{m(x_i; \hat{\mathbf{w}}_l)}\right) - \left(\log \frac{m(x_i; \mathbf{w}_k)}{m(x_i; \hat{\mathbf{w}}_k)}\right). \tag{15}$$

$$\mathcal{L}_{\mathcal{P}} = \mathbb{E}_{(x_i, l, k)\,:\, l \in \mathcal{P},\, y_{il}=+1,\, k \in S_{il}} \left[-\log \sigma\left(\beta \cdot h_{\mathbf{w}}(x_i, l, k)\right)\right]. \tag{16}$$

Minimizing $\mathcal{L}_{\mathcal{P}}$ makes $h_{\mathbf{w}}$ large and positive, so the model separates true positive privileged labels from their confusing negatives, relative to the reference.

Keeping Non-Privileged Performance Near the Reference

- For non-privileged labels $j \in \bar{\mathcal{P}}$, the goal is to keep performance from dropping relative to the reference $\hat{\mathbf{w}}_j$ while the model focuses on privileged labels.
- Base BCE loss for label j and instance x_i:

$$loss(\mathbf{w}_j) = -y_{ij} \log m(x_i; \mathbf{w}_j) - (1 - y_{ij}) \log(1 - m(x_i; \mathbf{w}_j)), \tag{17}$$

$$loss(\hat{\mathbf{w}}_j) = -y_{ij} \log m(x_i; \hat{\mathbf{w}}_j) - (1 - y_{ij}) \log(1 - m(x_i; \hat{\mathbf{w}}_j)). \tag{18}$$

- A hinge penalizes the model only when the increase in loss over the reference exceeds a slack $\epsilon \geq 0$:

$$\mathcal{L}_{\bar{\mathcal{P}}} = \mathbb{E}_{(x_i, \{y_{ij} : j \in \bar{\mathcal{P}}\})} \left[L_h(\mathbf{w}_j, \hat{\mathbf{w}}_j; \epsilon) \right], \tag{19}$$

$$L_h(\mathbf{w}_j, \hat{\mathbf{w}}_j; \epsilon) = \max\left(0,\ loss(\mathbf{w}_j) - loss(\hat{\mathbf{w}}_j) - \epsilon\right). \tag{20}$$

Part 3: FairPO

Group-Robust Balancing and Algorithm

Balancing the Two Group Losses

- The two losses pull against each other, and a fixed weighted sum does not adapt during training.
- We treat the two label groups as the groups in Group Robust Preference Optimization (GRPO = Group Robust Preference Optimization)[a] and write a minimax objective:

$$\min_{\{\mathbf{w}_t\}} \max_{\alpha \in \Delta} \left[\alpha_{\mathcal{P}} \mathcal{L}_{\mathcal{P}} + \alpha_{\bar{\mathcal{P}}} \mathcal{L}_{\bar{\mathcal{P}}}\right]. \tag{21}$$

- The weights $\alpha = (\alpha_{\mathcal{P}}, \alpha_{\bar{\mathcal{P}}})$ live on the simplex Δ. The inner step puts more weight on whichever group currently has the higher loss; the outer step updates the model under that weighting.

[a] Ramesh et al., *Group Robust Preference Optimization in Reward-free RLHF*.

Training Procedure

- Solved by alternating mirror ascent on α and mirror descent on $\mathbf{w}$.
- For stability, α is updated from each group's loss relative to its running average, since the two losses can be on different scales.

Algorithm 2 FairPO Algorithm for Multi-Label Classification (DPO-inspired)

Initialize: model parameters $\{\mathbf{w}_t^{(0)} \in \mathbb{R}^d | \forall t \in \mathcal{T}\}$ (e.g., copy $\{\hat{\mathbf{w}}_t | \forall t \in \mathcal{T}\}$), group weights $\alpha_{\mathcal{P}}^{(0)} \leftarrow 0.5, \alpha_{\bar{\mathcal{P}}}^{(0)} \leftarrow 0.5$.
Choose: learning rates $\eta_{\mathbf{w}}, \eta_\alpha$, DPO hyperparameter β, reference parameters $\{\hat{\mathbf{w}}_t | \forall t \in \mathcal{T}\}$, constraint slack ϵ.
for $s = 0$ to S (MaxIterations) **do**
  Sample an example: $(x_i, [y_{i1}, \dots, y_{iT}]) \in \mathcal{D} \sim p_{\mathcal{D}}(.)$.
  Initialize group losses for this step: $\mathcal{L}_{\mathcal{P}}^{(s)} \leftarrow 0, \mathcal{L}_{\bar{\mathcal{P}}}^{(s)} \leftarrow 0$.
  Initialize gradients: $g_{\mathcal{P}}^t \leftarrow \vec{0}, g_{\bar{\mathcal{P}}}^t \leftarrow \vec{0} \quad \forall t \in \mathcal{T}$.
  Forward pass: $m(x_i; \mathbf{w}_t^{(s)}) \leftarrow \sigma({\mathbf{w}_t^{(s)}}^T \mathbf{z}_i)$ where $\mathbf{z}_i \leftarrow \pi_\theta(x_i) \quad \forall t \in \mathcal{T}$.
  Sample a label: $r \in \mathcal{T} \sim Uniform(\frac{1}{|\mathcal{T}|})$.
  if $r \in \mathcal{P}$ **then** ▷ Handle privileged label
    $l \leftarrow r$
    $S_{il}^{\text{neg}} \leftarrow \emptyset, S_{il}^{\text{pos}} \leftarrow \emptyset$
    if $y_{il} = +1$ **then** ▷ True Positive case for privileged label l
      $S_{il}^{\text{neg}} \leftarrow \{k \in \mathcal{T} \mid y_{ik} = 0 \text{ and } m(x_i; \mathbf{w}_k^{(s)}) \geq m(x_i; \mathbf{w}_l^{(s)})\}$
      $S_{il} \leftarrow S_{il}^{\text{neg}}$
    else if $y_{il} = 0$ **then** ▷ True Negative case for privileged label l
      $S_{il}^{\text{pos}} \leftarrow \{k \in \mathcal{T} \mid y_{ik} = +1 \text{ and } m(x_i; \mathbf{w}_k^{(s)}) \leq m(x_i; \mathbf{w}_l^{(s)})\}$
      $S_{il} \leftarrow S_{il}^{\text{pos}}$
    end if
    if $S_{il} \neq \emptyset$ **then** ▷ Confusing examples exist, use DPO-inspired loss
      Sample $k \in S_{il} \sim Uniform(\frac{1}{|S_{il}|})$
      if $y_{il} = +1$ **then** ▷ DPO for True Positive l vs Confusing Negative k
        $h_{\mathbf{w}^{(s)}}(x_i, l, k) \leftarrow \left(\log \frac{m(x_i; \mathbf{w}_l^{(s)})}{m(x_i; \hat{\mathbf{w}}_l)}\right) - \left(\log \frac{m(x_i; \mathbf{w}_k^{(s)})}{m(x_i; \hat{\mathbf{w}}_k)}\right)$.
        $\mathcal{L}_{\text{pref}} \leftarrow -\log \sigma\left(\beta \cdot h_{\mathbf{w}^{(s)}}(x_i, l, k)\right)$
      else if $y_{il} = 0$ **then** ▷ DPO for True Negative l vs Confusing Positive k
        $h_{\mathbf{w}^{(s)}}(x_i, k, l) \leftarrow \left(\log \frac{m(x_i; \mathbf{w}_k^{(s)})}{m(x_i; \hat{\mathbf{w}}_k)}\right) - \left(\log \frac{m(x_i; \mathbf{w}_l^{(s)})}{m(x_i; \hat{\mathbf{w}}_l)}\right)$.
        $\mathcal{L}_{\text{pref}} \leftarrow -\log \sigma\left(\beta \cdot h_{\mathbf{w}^{(s)}}(x_i, k, l)\right)$
      end if
      $\mathcal{L}_{\mathcal{P}}^{(s)} \leftarrow \mathcal{L}_{\text{pref}}$
      $g_{\mathcal{P}}^t \leftarrow g_{\mathcal{P}}^t + \nabla_{\mathbf{w}_t} \mathcal{L}_{\text{pref}}|_{\mathbf{w}_t^{(s)}} \quad \forall t \in \mathcal{T}$.
    else ▷ No confusing examples, use BCE loss for privileged label l
      $\mathcal{L}_{\text{BCE}} \leftarrow -[y_{il} \log m(x_i; \mathbf{w}_l^{(s)}) + (1 - y_{il}) \log(1 - m(x_i; \mathbf{w}_l^{(s)}))]$
      $\mathcal{L}_{\mathcal{P}}^{(s)} \leftarrow \mathcal{L}_{\text{BCE}}$
      $g_{\mathcal{P}}^t \leftarrow g_{\mathcal{P}}^t + \nabla_{\mathbf{w}_t} \mathcal{L}_{\text{BCE}}|_{\mathbf{w}_t^{(s)}} \quad \forall t \in \mathcal{T}$.
    end if
  else if $r \in \bar{\mathcal{P}}$ **then** ▷ Handle non-privileged label
    $j \leftarrow r$
    $\ell(\mathbf{w}_j^{(s)}) \leftarrow -[y_{ij} \log(m(x_i; \mathbf{w}_j^{(s)})) + (1 - y_{ij}) \log(1 - m(x_i; \mathbf{w}_j^{(s)}))]$
    $\ell(\hat{\mathbf{w}}_j) \leftarrow -[y_{ij} \log(m(x_i; \hat{\mathbf{w}}_j)) + (1 - y_{ij}) \log(1 - m(x_i; \hat{\mathbf{w}}_j))]$
    $\mathcal{L}_{\bar{\mathcal{P}}}^{(s)} \leftarrow \max\left(0, \ell(\mathbf{w}_j^{(s)}) - \ell(\hat{\mathbf{w}}_j) - \epsilon\right)$
    $g_{\bar{\mathcal{P}}}^t \leftarrow g_{\bar{\mathcal{P}}}^t + \nabla_{\mathbf{w}_t} \mathcal{L}_{\bar{\mathcal{P}}}^{(s)}|_{\mathbf{w}_t^{(s)}} \quad \forall t \in \mathcal{T}$.
  end if
  $\alpha_{\mathcal{P}}^{(s+1)} \leftarrow \alpha_{\mathcal{P}}^{(s)} \exp(\eta_\alpha \mathcal{L}_{\mathcal{P},scaled}^{(s)})$ and $\alpha_{\bar{\mathcal{P}}}^{(s+1)} \leftarrow \alpha_{\bar{\mathcal{P}}}^{(s)} \exp(\eta_\alpha \mathcal{L}_{\bar{\mathcal{P}},scaled}^{(s)})$ ▷ Mirror ascent
  $Z \leftarrow \alpha_{\mathcal{P}}^{(s+1)} + \alpha_{\bar{\mathcal{P}}}^{(s+1)}$
  $\alpha_{\mathcal{P}}^{(s+1)} \leftarrow \frac{\alpha_{\mathcal{P}}^{(s+1)}}{Z}$ and $\alpha_{\bar{\mathcal{P}}}^{(s+1)} \leftarrow \frac{\alpha_{\bar{\mathcal{P}}}^{(s+1)}}{Z}$ ▷ Weight normalization
  $\mathbf{w}_t^{(s+1)} \leftarrow \mathbf{w}_t^{(s)} - \eta_{\mathbf{w}}(\alpha_{\mathcal{P}}^{(s+1)} g_{\mathcal{P}}^t + \alpha_{\bar{\mathcal{P}}}^{(s+1)} g_{\bar{\mathcal{P}}}^t) \quad \forall t \in \mathcal{T}$ ▷ Mirror descent
end for
return $\{\mathbf{w}_t^{(S)} | \forall t \in \mathcal{T}\}$

SimPO-Based Privileged Loss

SimPO[a] is reference-free and adds a target margin γ. Applied here, it compares the model's confidence on the true positive label l against the confusing negative k:

$$\mathcal{L}_{\mathcal{P}}^{\text{SimPO}} = \mathbb{E}_{(x_i,l,k)\,:\,l\in\mathcal{P},\, y_{il}=+1,\, k\in S_{il}} \left[-\log\sigma\left(\beta\left(\log\frac{m(x_i;\mathbf{w}_l)}{m(x_i;\mathbf{w}_k)}\right)-\gamma\right)\right]. \quad (22)$$

[a] Meng, Xia, and Chen, "SimPO: Simple Preference Optimization with a Reference-Free Reward".

CPO-Based Privileged Loss

CPO[a] combines a margin-free preference term with an NLL regularizer on the true positive privileged label:

$$L_{\mathcal{P}}^{\text{CPO-prefer}} = \mathbb{E}_{(x_i,l,k)}\left[-\log\sigma\left(\beta\log\frac{m(x_i;\mathbf{w}_l)}{m(x_i;\mathbf{w}_k)}\right)\right], \quad L_{\mathcal{P}}^{\text{CPO-NLL}} = \mathbb{E}_{(x_i,l)}[-\log m(x_i;\mathbf{w}_l)]. \quad (23)$$

$$\mathcal{L}_{\mathcal{P}}^{\text{CPO}} = L_{\mathcal{P}}^{\text{CPO-prefer}} + \lambda\, L_{\mathcal{P}}^{\text{CPO-NLL}}. \quad (24)$$

Each variant replaces only the preference term inside the privileged loss; the balancing and non-privileged handling stay the same.

Part 3: FairPO

Results: MS-COCO and NUS-WIDE

MS-COCO ($\mathcal{P}$ = 20% least frequent labels, $\bar{\mathcal{P}}$ = remaining 80%)

Method	mAP		Sample F1		Accuracy		EMR		ΔmAP($\mathcal{P}$)
	$\mathcal{P}$	$\bar{\mathcal{P}}$	$\mathcal{P}$	$\bar{\mathcal{P}}$	$\mathcal{P}$	$\bar{\mathcal{P}}$	$\mathcal{P}$	$\bar{\mathcal{P}}$	
BCE SFT	86.32	90.65	61.43	64.89	94.89	98.12	35.78	36.98	Ref
BCE SFT + RW	87.25	89.85	62.68	64.11	95.95	97.93	47.43	33.81	+0.93
GDRO + BCE	87.92	90.41	62.31	64.75	95.72	98.05	46.12	35.16	+1.60
FairPO-DPO	88.02	89.97	63.45	63.65	97.89	98.04	62.19	35.12	+1.70
FairPO-SimPO	87.67	88.76	62.34	63.12	95.69	97.78	45.32	32.34	+1.35
FairPO-CPO	89.76	90.34	64.01	64.32	98.03	98.06	65.43	35.23	+3.44

NUS-WIDE ($\mathcal{P}$ = 20% least frequent labels, $\bar{\mathcal{P}}$ = remaining 80%)

Method	mAP		Sample F1		Accuracy		EMR		ΔmAP($\mathcal{P}$)
	$\mathcal{P}$	$\bar{\mathcal{P}}$	$\mathcal{P}$	$\bar{\mathcal{P}}$	$\mathcal{P}$	$\bar{\mathcal{P}}$	$\mathcal{P}$	$\bar{\mathcal{P}}$	
BCE SFT	63.53	70.24	48.12	55.83	91.51	95.22	19.32	11.56	Ref
BCE SFT + RW	65.12	69.14	49.51	54.73	92.33	94.81	21.23	10.32	+1.59
GDRO + BCE	64.84	69.91	49.13	55.62	92.11	95.13	21.02	11.34	+1.31
FairPO-DPO	66.34	69.05	51.71	54.52	93.92	95.04	27.91	11.21	+2.81
FairPO-SimPO	64.11	68.03	48.82	53.81	91.94	94.52	20.18	10.19	+0.58

Summary of Findings

- FairPO addresses the performance gap in MLC, where models do poorly on infrequent but important labels.
- It combines a conditional, DPO-style preference loss that improves privileged (rare) labels with a constrained loss that keeps non-privileged labels from degrading, balanced by group-robust optimization.
- Across MS-COCO and NUS-WIDE, FairPO-CPO gives the largest privileged-group gain ($\Delta\mathrm{mAP}(\mathcal{P})$ of $+3.44$ and $+3.59$) while keeping non-privileged metrics close to the BCE reference.
- Fixing the group weights at $0.5/0.5$ instead of the adaptive balancing lowers the privileged gain (to $+2.16$ on MS-COCO), so the balancing step contributes to the result.

Publication

FairPO: Robust Preference Optimization for Fair Multi-Label Learning.[a] Accepted at the OPT workshop, NeurIPS 2025.

[a] Mondal, Chanda, et al., *FairPO: Robust Preference Optimization for Fair Multi-Label Learning*.

Conclusion

Summary, limitations, and future work

Conclusion

Main Findings

One Result per Study

- Part 1: editing language-specific neurons in Llama 3.1 and Mistral Nemo, at test time or through targeted fine-tuning, did not improve cross-lingual transfer on XNLI and XQuAD, and several edits lowered performance.
- Part 2: GRPO moved accuracy very little across seven MGSM languages; the English-leaning middle of the stack was present from the start and did not change with training, and the one language it did not improve (Telugu) was the one with the least tokeniser coverage.
- Part 3: FairPO improved the rare privileged labels on MS-COCO and NUS-WIDE while keeping the frequent labels close to the reference, with each part of the framework contributing to the result.

What the Three Studies Point To

- Acting on a small, localized set of internal components is not a reliable way to change cross-lingual behaviour. This held for the neuron edits in Part 1 and is consistent with Part 2, where the localized mechanism the collapse hypothesis points to was not what GRPO acted on.
- The training objective is the more effective place to act, but only where the training signal reaches the model at a fine enough level: GRPO helped a language only with enough tokeniser resolution, and FairPO helped because its objective acted directly on the label pairs that needed separating.
- A method that targets a chosen part of the label space and protects the rest can improve the targeted part without measurable harm to the rest, as FairPO showed.

Limitations

- Part 1 looks only at the MLP layers, uses two models and two tasks, and relies on Wikipedia statistics for the test-time interventions.
- Part 2 uses one model family and one task, fine-tunes a low-rank adapter on the attention projections only, and depends on a logit-lens readout and a judge-based reward, both imperfect; the coverage counts are approximate.
- Part 3 uses a frequency-based label partition on two image benchmarks, depends on a confusing set that changes during training, and needs care in tuning the group-robust balancing.

Future Work

- Cross-lingual transfer: since language is spread across the network, look at interventions on larger parts of the model, or on the training data and objective, rather than single neurons.
- Tokeniser coverage: change the tokeniser or add tokens for a low-coverage language and check whether GRPO can then improve it, to test whether coverage is the cause or only a correlate.
- Broader testing: run the same measurements on other model families, reasoning tasks, and more languages to see how far the per-layer pattern and the coverage effect hold.
- Extending FairPO: apply it to other sources of disparity such as domain-set label importance, and to generation tasks where the groups are defined over attributes, and study the stability of the confusing set and the convergence of the minimax objective.

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